

Reasoning in Governed LLM Agents: A Functional-Analogy Account of Observation, Reflection, and Planning

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Abstract

Claims that software agents built on large language models (LLMs) can “observe,” “reflect,” and “plan” are met with justified skepticism: on their face they either anthropomorphise a statistical process or trade on cognitive vocabulary the systems have not earned. This paper argues for a position narrower than the popular claim and stronger than its dismissal. We proceed in three steps. First, we establish from cognitive science that observation, reflection, and planning are, in humans, generative and constructive inferential acts rather than retrievals of stored answers, and that this generativity, not any particular substrate, is what the three verbs denote. Second, we impose a discipline the neuroscience demands: because the language network is dissociable from the brain’s reasoning networks, fluency cannot be taken as evidence of reasoning, and functional competence must be demonstrated separately from formal linguistic competence. Third, we show that a bare autoregressive model performs a defensible *functional analogue* of the generative component of each cognitive function, strongest over already-structured inputs, but that it demonstrably lacks referential grounding, faithful access to its own computation, and robustness under distribution shift. Our central claim is that these three deficits are not a refutation but a specification. A deterministic process harness of the kind used in Sequential Decision Analytics (developed in a companion paper) supplies precisely what the model lacks: it grounds the model’s output in recorded state and objective functions, it declines to rely on the model’s own reasoning trace when making decisions, and it gates action behind thresholds, constraints, and corroboration. Reasoning, on this account, is a distributed property of the composite system. We give an evidence-graded mapping of each cognitive function to its machine analogue and to the harness requirement it induces, and we identify the conditions under which the account strengthens or fails.

Keywords: large language models, cognitive architecture, active inference, metacognition, prospection, symbol grounding, chain-of-thought faithfulness, agentic AI, sequential decision analytics, governed autonomy

1 Introduction

An operational AI agent that reads plant telemetry, writes an interpretation of the current state, synthesises that interpretation against its memory, and produces a plan of action is routinely described as observing, reflecting, and planning. The description invites a reasonable objection. Software does not observe; it computes over inputs. To say it reflects is either to anthropomorphise a lookup or to borrow the connotations of a human capacity the system does not possess.

There are two common answers to this objection and both are wrong. The first overclaims: the agent thinks, understands, and reasons in the way a person does. The second dismisses: the system is autocomplete, a stochastic parrot, and nothing cognitive is occurring. The evidence supports neither. The overclaim collides with a large body of work on the limits of language models; the dismissal collides with an equally large body of work on what those models demonstrably do.

This paper defends the intermediate position and, more importantly, identifies the mechanism that makes it hold. The position is that an LLM performs a genuine *functional analogue* of the generative core of observation, reflection, and planning, and that the cognitive validity of the composite agent depends on a deterministic process harness that compensates for the model’s specific and well-documented deficits. The contribution is threefold:

1. A synthesis, drawn from cognitive neuroscience and cognitive psychology, that recasts the three target functions as generative inference and thereby dissolves the “retrieval versus cognition” framing on which the objection rests (Section 2).
2. A methodological discipline, taken from the neural dissociation between language and thought, that forbids inferring reasoning from fluency and requires functional competence to be earned separately (Sections 3 and 4).
3. A reframing of the standard skeptical arguments, grounding, faithfulness, and robustness, not as a refutation of agentic reasoning but as a requirements

specification, together with a demonstration that a governed-decision architecture satisfies each requirement (Sections 6 and 7).

We restrict scope deliberately. The decision-theoretic content of the harness, its formulation as a sequential decision problem with state variables, policies, and objective functions in the sense of Powell’s unified framework for Sequential Decision Analytics [30], is treated in a companion paper. Here the harness appears only insofar as it bears on the cognitive question: whether, and in what sense, the composite system can be said to reason. Throughout, we grade claims as **[Established]**, **[Emerging]**, or **[Contested]** following the convergent assessment of several independent evidence reviews, and we anchor examples in a reference implementation, the XMPro MAGS ORPA loop (Observe, Reflect, Plan, Act), without treating the argument as specific to it.

2 Observation, Reflection, and Planning Are Generative, Not Retrieval

The objection quietly assumes that human observation, reflection, and planning are themselves lookups, so that a generative computer process could only be a poor imitation of them. Contemporary cognitive science rejects that assumption. Each of the three functions is, in the human case, an active and constructive inferential act.

Observation as active inference. **[Established]** The dominant framework for perception, predictive processing, models the brain as a hierarchical generative system that predicts its sensory input from prior expectations and revises those predictions against residual error, propagating only unexplained error upward [8, 13]. Perception is therefore a controlled construction that uses learned priors to stabilise noisy and ambiguous input, not a passive transcription of the world. The stronger metaphysical readings of this framework, including the “controlled hallucination” formulation [32], remain **[Contested]**, and predictive processing is best described as a leading and influential framework rather than a closed question. The claim we require is the weaker and well-supported one: perceiving is inferring a structured interpretation of what is happening.

Reflection as constructive metacognition. **[Established]** Reflective reasoning is the slow, serial, effortful mode that monitors and revises fast automatic processing [10], operating through the monitoring-and-control loops of metacognition [27]. Its best-evidenced generative mechanism is the self-explanation effect: prompting a learner to generate explanations to themselves improves comprehension and error detection, because producing explicit intermediate statements forces conceptual gaps and contradictions into view [7]. A meta-analysis drawing 69 effect sizes from 64 research reports estimates a moderate effect, with an overall random-effects weighted mean of

$g = 0.55$ [6]. Reflection is thus the generation of an intermediate representation for the purpose of checking and revision, not the recall of a stored conclusion. Inner speech scaffolds this process but, on the careful reading of the evidence, facilitates rather than constitutes it [1].

Planning as prospective simulation. **[Established]** Planning is mental time travel: the projection of the agent into non-current states to simulate and evaluate candidate futures before acting [34]. It relies on hierarchical goal representations maintained by prefrontal cognitive control [24] and on the construction and manipulation of mental models of possibilities [16]. Under time pressure, expertise bypasses exhaustive option comparison: the recognition-primed decision model shows that experienced operators recognise a situation as typical and mentally simulate a single promising course of action to test it before committing [18]. Planning is therefore generative construction and evaluation of possible trajectories.

A methodological caution belongs here. The ORPA loop resembles the Observe-Orient-Decide-Act (OODA) loop, but OODA is practitioner doctrine rather than peer-reviewed cognitive theory and should not be cited as scientific support. The scientific anchors for the three verbs are the ones above: predictive processing, metacognition and self-explanation, and prospection with recognition-primed decision-making.

The upshot of Section 2 is that the sharp line the objection draws, between a generative machine and a retrieval-free human faculty, does not exist. In the human case these faculties are themselves generative. That does not make the machine equivalent; it removes the categorical objection and turns the question into an empirical one about how far the analogy extends.

3 The Dissociation Constraint: Fluency Is Not Reasoning

Before extending the analogy to machines, we must impose a constraint that most popular treatments omit, and whose omission is the source of the overclaim. It is tempting to reason: these functions are often verbal; the model is fluent; therefore the model performs the functions. The neuroscience blocks this inference directly.

The human language network is dissociable from the systems that carry out reasoning. High-resolution functional imaging and clinical lesion studies show that the frontotemporal language network is selective for linguistic form and remains largely inactive during arithmetic, logical deduction, and executive reasoning, which recruit the domain-general frontoparietal Multiple Demand network [11, 12]. Individuals with severe global aphasia, whose linguistic capacity is profoundly impaired, retain the ability to solve logic and arithmetic problems, reason causally, and attribute mental states to others. **[Established]** Language

is an augmentative medium for higher cognition, not its substrate.

Mahowald et al. [23] carry this distinction into the evaluation of language models with the framing that should govern the entire discussion: *formal* linguistic competence, mastery of the structure of language, is separable from *functional* linguistic competence, the use of language to accomplish goals in the world. LLMs have substantially mastered the formal dimension while remaining uneven and brittle on the functional one. **[Established as a framework]**

The methodological consequence is decisive. Inferring reasoning from fluent output is precisely the inference the dissociation warns against. A claim that an agent observes or reflects cannot rest on the fluency or plausibility of its generated text. Functional competence must be demonstrated by other means. This constraint does two things for the argument. It disqualifies naive capability claims, and it foreshadows the resolution: if fluency cannot certify reasoning, something outside the generator must. That is the role the process harness will play.

4 Token Generation as a Functional Analogue

Within the discipline of Section 3, the positive claim can be stated precisely. In each phase of an agent’s loop, the language model performs a functional analogue of the *generative* component of the corresponding human function. The analogue is strongest where the input is already symbolically structured and where success depends on serial decomposition and explicit intermediate state rather than on raw sensation. This is the regime in which operational agents work.

Observe. The agent renders an input, together with retrieved context, into a structured interpretation of the current state. This is not raw perception and should not be described as such. It corresponds to the reportable, interpretive layer that expert operators run, what the naturalistic-decision literature calls model-based situation assessment. In the reference implementation, the observation phase issues a single model call that produces an analysis of the input against retrieved knowledge context; the analogy is legitimate precisely because the input is structured telemetry and events rather than sensory transduction.

Reflect. The agent synthesises recent observations and prior reflections into a higher-order representation. This is the closest analogy in the loop. Generating intermediate tokens measurably improves multi-step performance in models, for reasons that parallel the self-explanation effect: chain-of-thought prompting and scratchpad generation supply serial working space in which a problem is decomposed [28,36]. **[Established for the performance effect; Contested as “reasoning” in the full sense.]** The gains can be large and are not merely presentational; for

example, chain-of-thought prompting substantially raised multi-step arithmetic performance for large models relative to standard prompting [36]. The reference implementation operationalises this as a memory-conditioned synthesis whose product is written back as a new memory.

Plan. The agent produces a structured plan and decomposes it into tasks. This maps to prospective simulation and hierarchical goal decomposition. In the reference implementation the model emits a formal plan representation and a task breakdown; the model performs the generative half of planning, the construction of candidate structured trajectories.

Independent mechanistic evidence indicates that sequence prediction can induce internal structure rather than only surface statistics. A model trained solely to predict legal Othello moves develops an internal, linearly encoded, and causally manipulable representation of the board state [20, 26]. **[Emerging.]** This is an existence proof in a closed, fully specified domain and must not be over-extended to open-world natural language, but it establishes that next-token training can yield usable latent world structure.

5 The Structured-Data Corollary

Operational agents reason over telemetry, measures, and process state rather than over conversational text. The corollary that licenses this is that any sufficiently structured and predictable sequence can be tokenised and modelled by the same predictive machinery, and it is empirically supported rather than merely asserted. **[Established for the engineering claim; Contested for any inference to “meaning.”]**

Protein language models trained by masked or autoregressive prediction over amino-acid sequences recover three-dimensional structure and function from sequence alone [21,31]. Time-series foundation models tokenise numeric telemetry, by patching or by scaling-and-quantisation into a fixed vocabulary, and achieve strong zero-shot forecasting with standard transformer decoders trained under cross-entropy loss [3,9]. Industrial process state is a structured, predictable language in this sense, and the transfer effects observed for text recur over telemetry.

The boundary of the corollary is as important as its content and it anticipates Section 6. These models track the intrinsic predictability of the process. They are weakest at regime shifts, rare faults, and genuine novelty, and small tracking errors compound into physically impossible states over long horizons. These are exactly the conditions that matter most in a safety-relevant operating environment. Prediction over structure is a real capability whose failure region coincides with the region of greatest operational consequence.

6 The Skeptical Case as a Requirements Specification

The strongest scientific objections to identifying token generation with observation, reflection, and planning are not answered by Sections 4 and 5, and we do not soft-pedal them. They are three, and each is independently decisive against the *strong* (identity) thesis. The reframing this paper offers is to read them not as a verdict but as a specification of what a system must add around a generator.

Grounding. [Established as a problem; Contested whether LLMs escape it.] The symbol grounding problem holds that symbols manipulated purely by their form, without connection to non-symbolic referents, have no intrinsic meaning; their meaning is parasitic on the human interpreters who read them [15]. A model that emits the tokens for “high system temperature” has no somatic concept of heat, pressure, or hazard; it predicts a sequence that co-occurred with those concepts in training. The “stochastic parrots” critique makes the parallel point about fluent recombination of form without reference [4, 5]. A balanced survey concludes that neither “understands fully” nor “understands nothing” fits the evidence, and that the field lacks adequate vocabulary for the middle case [25]. *Requirement induced: meaning that drives action must be anchored outside the generator, in representations connected to the real process.*

Faithfulness. [Established.] This is the objection with the greatest force for any system that acts on its outputs. The visible reasoning trace is not a reliable record of the model’s actual computation. Turpin et al. [35] show that introducing a hidden bias, such as systematically making the correct answer a fixed option, substantially shifts the model’s prediction while its generated explanation rationalises the biased choice and never mentions the bias. Chain-of-thought can be post-hoc rationalisation. The artefact that most resembles reflection therefore cannot be trusted as a transcript of the reason for a decision. *Requirement induced: decisions must not be adjudicated by the model’s own generated rationale.*

Robustness. [Established/Emerging.] Functional competence is brittle under distribution shift and adversarial paraphrase, and models confabulate confidently at the edges of their training distribution, which in operational terms are the fault and novelty regimes (Section 5). *Requirement induced: action must be gated by checks that do not depend on the generator’s confidence.*

Together these refute the identity thesis. A bare language model is not observing, reflecting, or planning in the grounded, self-transparent, and robust sense a human is. To assert otherwise, particularly in a control setting where an unfaithful rationale or a confident out-of-distribution extrapolation is a hazard, is not merely imprecise but unsafe. What the objections do not refute is a *composite* system that is engineered to the requirements they induce.

7 The Process Harness as the Resolving Mechanism

The requirements derived in Section 6, external grounding, non-reliance on the generated rationale, and confidence-independent gating, define what a system must add to a generator before the cognitive verbs are earned. A deterministic process harness of the kind used in governed multi-agent architectures satisfies each. We describe the mechanism generally and note where the reference implementation realises it.

Grounding by external anchoring. The harness does not let the model’s words stand as meaning. The significance of an observation is computed deterministically from importance and surprise, and these quantities are themselves defined by cognitive-science-derived formulations, drawing on prospect theory, information theory, and Weber-Fechner scaling, rather than by the model’s assertion. Candidate plans are scored by an objective function defined over real measures and constraints, with aggregation strategies that can order priorities lexicographically so that, for example, safety precedes throughput. The model proposes an interpretation; whatever drives a decision is anchored in quantities the harness owns and that are connected to the process. This is the operational answer to the grounding objection: reference is supplied by the layer in contact with the world.

Faithfulness by non-reliance. The correct engineering response to the unfaithfulness result [35] is to stop treating the reasoning trace as the reason. The harness does exactly this. Its decision records retain the raw model input and output for audit rather than relying on them, and every consequential decision, whether to reflect, whether to plan, which plan to adopt, is a deterministic function of recorded state: a significance threshold for reflection, an adaptation detector with an explicit guard for replanning, objective and utility scoring for plan selection, and a corroboration check across agents before commitment. The generated explanation is logged as one further input, tagged as what it is, and never adjudicates the outcome. The architecture is, in effect, already built to the faithfulness literature.

Robustness by gated action. The failure regimes identified in Sections 5 and 6, regime shifts and rare faults, are met by the same deterministic gating: thresholds, constraint checks, and multi-agent consensus corroboration precede any committed action, none of which depends on the generator’s self-reported confidence. A confident hallucination at the edge of distribution does not by itself produce an action.

The unifying stance is to bound the model as *exogenous information*, occupying the same position as a sensor reading: useful, fallible, and never authoritative on its own. One trusts a sensor to produce a value and still passes that value through logic before acting on it; the harness treats the model identically. This is neither the overclaim

nor the dismissal. It is the functional-analogy position with the boundary drawn exactly where the evidence in Section 6 requires it.

This yields the paper’s central claim. Reasoning in a governed LLM agent is not a property of the model and is not a property of the harness. It is a **distributed property of the composite**. The model contributes generative inference, which Section 2 shows is the core of what the three verbs denote and which Sections 4 and 5 show the model genuinely performs over structured inputs. The harness contributes grounding, faithfulness, and robustness, which Section 6 shows the model lacks and which are necessary conditions for the verbs to apply to a system that acts. Neither component reasons in the full sense; the composite satisfies the functional criteria.

8 A Formal Statement and Evidence-Graded Mapping

We state the defensible claim precisely, then map it.

A governed LLM agent observes, reflects, and plans in a functional, task-relative sense. The claim is warranted because (i) these functions are generative inferential acts rather than retrievals, (ii) the model performs the generative component over structured representations where the analogy is strongest, and (iii) a deterministic harness supplies the grounding, faithfulness, and robustness that the generative component provably lacks. The claim is not that the agent observes, reflects, or plans in the grounded, self-transparent, autonomous sense characteristic of human cognition. The functional criteria are met by the composite; they are met by neither component alone.

The mapping in Table 1 relates each function to its biological mechanism, its machine analogue, the strength of the analogy on current evidence, the deficit in the bare model, and the harness requirement the skeptical case induces.

The grading is deliberately conservative and matches the convergent reading of the underlying literature: the generative analogy is real and, for reflection, quite close on the performance dimension, while every row carries a deficit that is closed not by a better prompt but by a mechanism outside the model.

9 Related Work and Positioning

The argument sits at the intersection of three research programmes, and its contribution is clearest when set against each.

Cognitive architectures. Classical symbolic architectures such as Soar [19] and ACT-R [2] model cognition as

a control cycle over structured representations: perceive, match production rules, select, and act, over declarative and procedural memory. ORPA echoes that cycle, but the substrate is inverted. Where a classical architecture hand-builds the productions that generate candidate interpretations and actions, the present account delegates that generative step to a learned model and retains a deterministic control layer for the selection and gating that the classical architectures located in their decision cycle. We therefore make no claim to a unified cognitive model; we claim a division of labour in which the generative substrate is learned and the governing policy is engineered.

LLM agent scaffolds. A large recent literature adds reasoning and acting structure around a language model: interleaving reasoning traces with tool actions (ReAct [37]), searching over intermediate thoughts (Tree of Thoughts [38]), iterative self-feedback (Self-Refine [22]), verbal self-reinforcement from reflection (Reflexion [33]), and long-running agents with an observation, reflection, and planning memory loop (Generative Agents [29]). This body of work establishes empirically that generated intermediate structure improves multi-step behaviour, which supports Section 4. The present account differs on where the decision lives. Most of these methods rely on the model to evaluate its own output through self-critique, self-refinement, or self-scored reflection. That is precisely the faculty the faithfulness literature shows to be unreliable [35]: a model’s self-report is not a dependable readout of the computation that produced its answer. Generative Agents is the closest prior work in shape, since it runs an observe-reflect-plan cycle over memory and even assigns an importance score to memories, but that importance score and those reflections are produced by the language model itself. The architecture described here removes the consequential judgement from the model and computes significance, plan value, and adaptation triggers deterministically from recorded state. The scaffolds make the generator more capable; the harness makes the decision trustworthy, and it does so by declining to trust the generator’s self-assessment.

Neuro-symbolic AI and generate-then-verify. The closest programme is the effort to couple neural generation with sound external checking. Neuro-symbolic AI broadly seeks to integrate learned and symbolic computation [14]. The nearest specific proposal is the LLM-Modulo framework [17], which argues that language models are effective generators of candidate plans but poor guarantors of them, and that planning competence must reside in external sound verifiers that accept or reject the model’s proposals. Our thesis agrees with this division and extends it in three ways. First, it grounds the division in cognitive science, explaining why the generative half is a legitimate functional analogue of the cognitive verbs rather than only a useful heuristic. Second, it generalises the scope from planning alone to the full observe-reflect-plan cycle, deriving a distinct requirement, grounding, faithfulness, or robustness, for each phase from the corresponding skeptical

Table 1: Mapping of each cognitive function to its biological mechanism, machine analogue, analogy strength, the deficit in the bare model, and the harness requirement that closes it.

Function	Biological mechanism	Machine analogue	Analogy strength	Deficit in the bare model	Harness requirement that closes it
Observe	Active inference; prior-driven construction of a structured interpretation [8, 13]	State interpretation generated from structured input plus retrieved context	<i>Partial</i> , stronger for pre-structured inputs	No sensorimotor grounding; open-loop over tokens	Significance and confidence scored deterministically; interpretation anchored to owned measures
Reflect	Metacognitive monitoring; self-explanation [6, 7]	Intermediate-token synthesis over memory (chain-of-thought, scratchpad)	<i>Partial to strong</i> for performance; weak for faithfulness	Generated trace is often post-hoc, unfaithful [35]	Decisions taken from recorded state, not from the trace; trace retained for audit only
Plan	Prospective simulation; hierarchical goal control [18, 24, 34]	Generated structured plan and task decomposition	<i>Partial</i> ; weak without an external evaluator	No endogenous goals; confabulation under novelty	Plans scored by objective/utility functions; action gated by constraints and consensus

argument. Third, it instantiates the verifier not as a plan validator but as a decision-analytics policy in the sense of Sequential Decision Analytics [30]: objective and utility functions over real measures, significance thresholds, and multi-agent corroboration. The contribution of this paper is therefore not the generate-then-verify pattern itself, which is established, but the cognitive-science account of why it earns the vocabulary of reasoning, together with the reframing of the skeptical case as the specification the governing policy is built to satisfy.

10 Limitations and Open Questions

Several limitations bound the account. First, it is a functional-analogy argument, not an identity argument, and it makes no claim about phenomenology, understanding in the referential sense, or autonomous goal formation; the steelman of the grounding objection [4, 15] stands against the strong thesis and we concede it. Second, the frameworks the argument leans on are frameworks: predictive processing and dual-process theory are influential but not closed, and their strong readings are contested. Third, the emergent-world-model evidence [20] is drawn from closed synthetic domains and does not license claims about open-world representation. Fourth, the structured-data corollary holds within the predictable regime and is expected to degrade precisely at the rare and novel events of greatest operational importance; the harness mitigates but does not eliminate this exposure.

The account also implies falsifiable conditions under which it would strengthen or weaken. If measures of chain-of-thought faithfulness, for instance causal-intervention agreement between stated reasoning and actual computation, were shown to be high and verifiable on a target domain, the reflection analogy would strengthen from partial toward strong and the non-reliance requirement could be relaxed. Conversely, if a deployed system were shown to make consequential decisions that in fact track

the model’s unfaithful trace rather than the deterministic policy, the central claim would fail for that system. And if interpretability work demonstrated causally manipulable world models in the open natural-language domain, the grounding objection would weaken materially. The account is therefore not merely a defensive hedge; it specifies what evidence would move it.

11 Conclusion

The question of whether an LLM agent can observe, reflect, and plan is usually argued at the level of the model, and at that level the honest answer is no in the strong sense and yes only in a hedged functional sense that critics rightly find unsatisfying. The argument changes when the unit of analysis is the composite system. Cognitive science establishes that the three functions are generative inference; the dissociation between language and thought forbids reading reasoning off fluency and requires functional competence to be earned separately; the capability literature shows the model performs the generative component over structured data; and the skeptical literature specifies, with unusual precision, the three things the model lacks. A deterministic process harness supplies exactly those three things. Reasoning, on this account, is a property distributed across a generator and its governance: the model generates, the harness grounds, refuses to trust the trace, and gates the act. The plain-language intuition that generation differs from retrieval, and that the task the model performs is distinct from the purpose the surrounding system holds, is a useful entry point. The cognitive science is what tells us why the intuition holds and, equally, where it stops.

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